Alim Yakub

Research Associate | Energy & Sustainability

alim.ykb@pm.me Singapore https://bukaymila.github.io/home Alim BMY

Summary

Research Engineer bridging sustainable thermal design and data science. Currently developing low-cost cold chain solutions via 3D prototyping, PCM experimentation, and Python analytics. Experienced in ML optimization, experimental design, and academic publishing across energy systems and engineering research.

Education

Newcastle University — Master of Science (MSc) • Distinction
Energy & Sustainability — Singapore • 2018 – 2019

Newcastle University — Bachelor of Science (BSc) • Hons (2:1)
Mechanical Design & Manufacturing Engineering — Singapore (SIT) • 2016 – 2018

Experience

Republic Polytechnic — Singapore
Research Staff — Jul 2025 – Present

- Designed and iterated cold box prototypes via CAD and FDM 3D printing for rapid design-test cycles. Conducted material trade-offs to select sustainable, low-cost insulation and structural materials.
- Designed and executed DOE (Design of Experiments) frameworks for shelf-life stability and PCM thermal characterization. Developed SOPs for all experimental protocols, ensuring reproducibility and lab safety.
- Built HTML/JavaScript and Python dashboards for data analysis/visualization.
- Supervised several Applied Science FYP teams for shelf-life studies (microbial, vitamin A&C, texture analysis, moisture analysis and water activity). Supervised Engineering FYP students for the design, material and in-house PCM testing for cold box prototype.

CNRS@CREATE — Singapore
Research Associate — Jan 2023 – Sep 2023

- Migrated a MATLAB optimization program to Python to replace the optimization controller of a grid-connected distributed energy system, implementing an adaptive, data-driven linear tree controller that reduced model footprint and enabled real-time deployment.

Surbana Jurong-NTU Corporate Lab — Singapore
Research Associate — Apr 2022 – Sep 2022

- Developed a Python toolkit combining ML-based efficiency prediction (regression models) with optimization of chiller sizing and daily operations, enabling data-driven decisions that balances cost, energy consumption and carbon lifecycle.

NTU Institute of Science & Technology for Humanity — Singapore
Research Associate — Nov 2020 – Apr 2022

- Developed a Python/JavaScript platform to collect, analyze, and visualize NISTH fellows' publications mapped against the UN Sustainable Development Goals, covering various publications across all SDGs.
- Designed and built the NISTH 2021 Annual Web Report, delivering an interactive online publication for stakeholders.
- Contributed to research report writing and generated new research topic ideas at the intersection of science, technology, and the humanities.
- Supervised an undergraduate FYP team building a social robot for elderly users to promote intergenerational connectivity, guiding the project to completion.

Skills

Python
NumPy, Pandas, Scikit-learn, Matplotlib, SciPy

Python machine learning
Regression, Decision Trees, Controller

3D CAD modelling
Autodesk Inventor, SolidWorks, Autodesk REVIT, IESVE

Data analytics and visualization
HTML/JavaScript, Plotly, D3

Engineering & Human-centered design
HMI, Sustainable design

Research Interests

Scientific and Social Aspects of Sustainability, Distributed Energy Resources, Design, Renewable Energy Generation, Robotics & Controls, Energy Storage

Languages

English

Mandarin

Malay

Publications

Decision Tree Variations and Online Tuning for Real-Time Control of a Building in a Two-Stage Management Strategy
DOI: 10.3390/en17112730

References

Rémy Rigo-Mariani
Research Scientist
remy.rigo-mariani@g2elab.grenoble-inp.fr
Centre national de la recherche scientifique (CNRS)

Michael Lau
Associate Professor
michael.lau@ncl.ac.uk
Newcastle University

Vanessa Evers
Director
vanessa.evers@ntu.edu.sg
NTU Institute of Science and Technology for Humanity